**Workbook Template**

**Project Name: News aggregation with category filter**

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## Problem Statement:

people find it hard to read news from many sources and choose only the news they want by category, so we need one website that collects news and shows it in organized categories.

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## Why does this problem exist?

News comes from website, apps, social media, and TV, so people see far more headlines than they can read or remember. Brain gets overloaded, hard to focus. Fake or low quality news spreads fast. Due to negative headlines, people feels anxious and start ignoring news without usefull information.

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## Who are the stakeholders?

Internal users:
1. Admin/platform owner
2. Content editor
3. Developer
4. Advertisers

External users:
1. Readers

**Expected system users:**

1. General readers
2. Students and learners
3. professionals
4. Mobile users
5. Researchers
6. Project evaluators

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**Objectives of the program:**

1. Sort news into clear categories(tech,sports etc.).
2. Let users filter news by category.
3. Show latest and relevant headlines first.
4. Provide a simple,clean, mobile-friendly UI.

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**Functional requirements:**

1. User registration & login: Users can create an account and login.
2. Display news: The system displays the latest news.
3. Category filter: Users can filter news by categories such as sports, technology, politics, business etc.
4. Search news: Users can search for news using keywords.
5. Read news: Users can open and read the full news article.
6. News management: Admin can add,update,and delete news.
7. Category management: Admin can add,edit,or delete the categories.
8. News sorting: News can be sorted by date or category.
9. Logout: Users/admin can securely logout.

**Non-functional requirements:**

1. Performance: News should load quickly.
2. Security: User and admin information should be protected.
3. Usability: The website should be simple and easy to use.
4. Reliability: The system should work consistently without frequent errors.
5. Maintainability: The code should be easy to modify and maintain.
6. Availability: The website should be accessible whenever the server is running.
7. Accuracy: Displayed news information should be stored correctly.

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**Expected outputs:**

1. A working website that shows news from many sources in one place.
2. News displayed as cards/list with title, source,date,and "Read more" link.
3. Category buttons(all tech,sports, business etc.)
4. Latest news appears first(sorted by date).
5. Simple,clean design that works on desktop and mobile.
6. Admin to store new articles in the database automatically.

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**Hands-on activity:**

1. Plan categories: Sports,tech,politics, business etc
2. Get news API key: Current news, NewsAPI.org (100 requests/day) etc.
3. Build UI: Navbar with category buttons + news feed area.
4. Fetch API data: Use fetch to get news.
5. Display news cards: Title,image, description, source link.
6. Add filter logic: Click category then reload filtered news.
7. Add search bar: Keyword-based search(optional).

8. Add loading/error states: Spinner,"no internet" message.
9. Make responsive: Works on mobile and desktop.
10. Deploy: Putting your project online so anyone can open it using a link.

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**MIS components:**

**1. People(user):**

Users/readers who view news.
Admin who manages news, categories, and users.

**2. Hardware:**

Computer/laptop, mobile phone, server, and networking devices.

**3. Software:**

Website/application for displaying news.

**4. Data:**

News title, description, image, source, date, author, and category.
Categories can include Sports, Politics, Technology, Education, Business, Entertainment, etc.

**5. Procedure/processes:**

Collect news -> store in database -> categorize news -> filter news -> display to users.
Admin can add, update, or delete news.

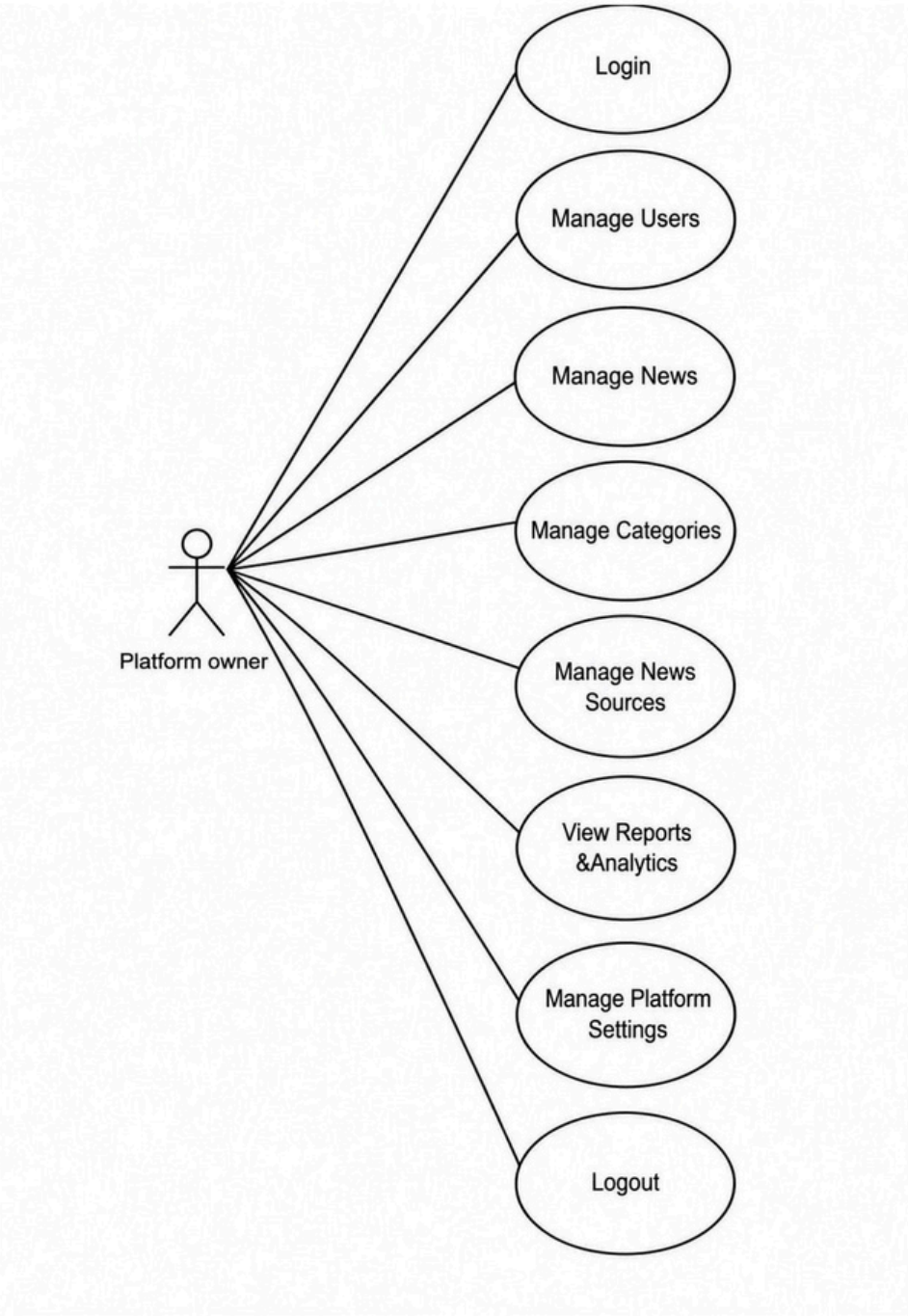
**6. Database:**

Store users, news articles, categories, and other information.
Helps retrieve news quickly based on the selected category.

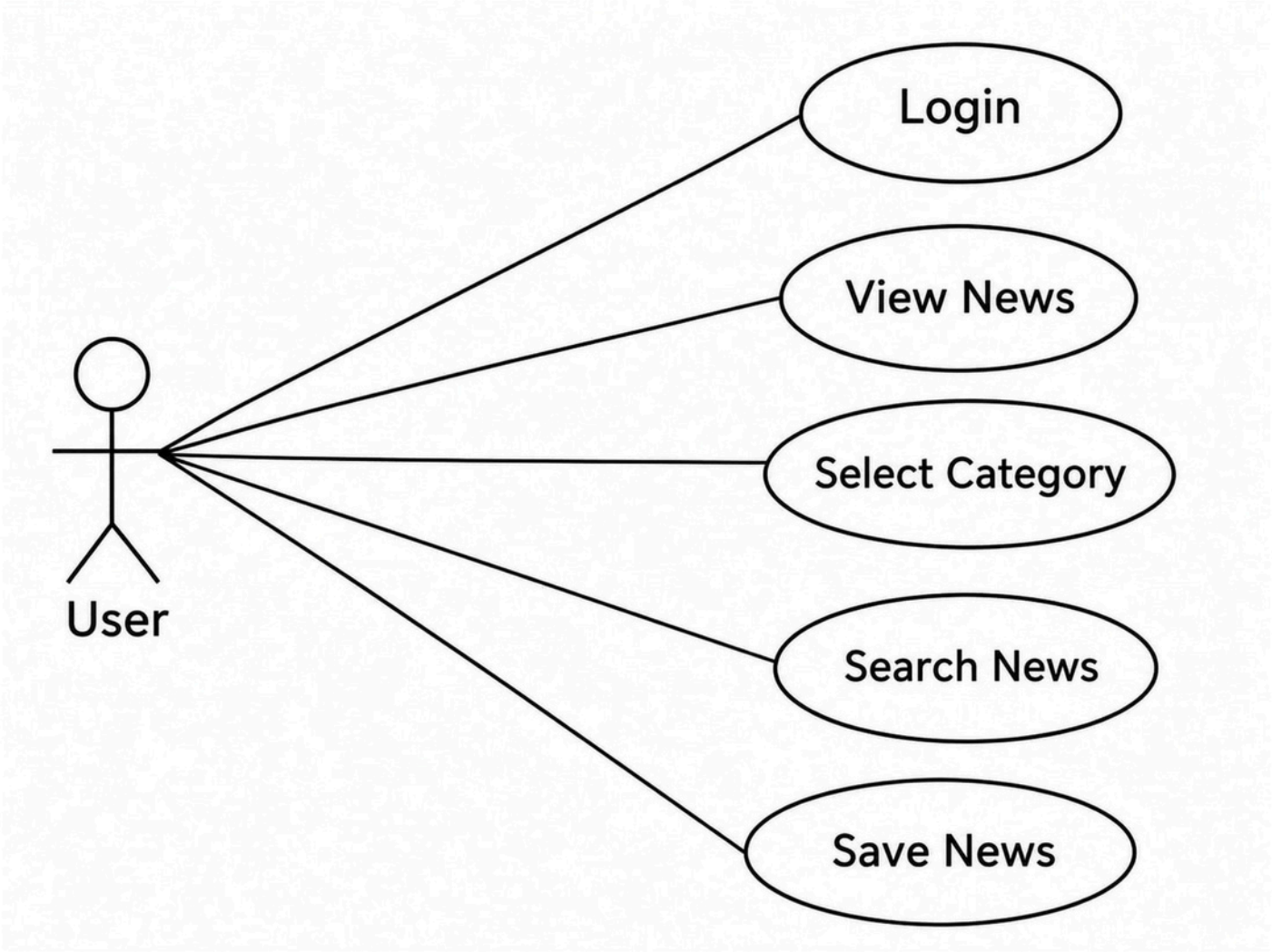
**7. Network:**

Internet/network connects the application with users and news sources.
Allows users to access updated news online.

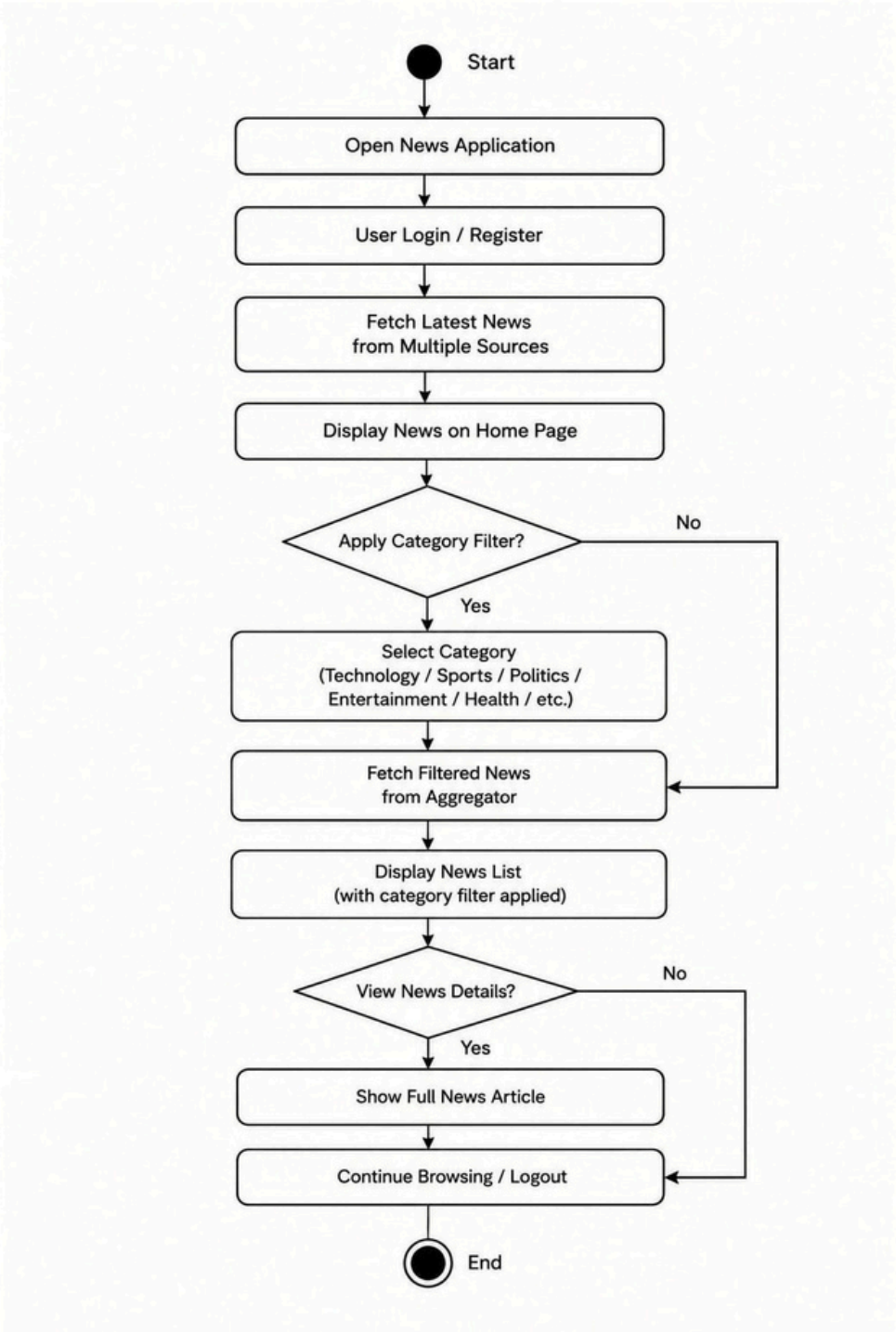
# Use Case Diagram of
# News aggregation with category filter



# Use Case Diagram - User



# Activity Diagram of
# News aggregation with category filter



# Types of Analysis

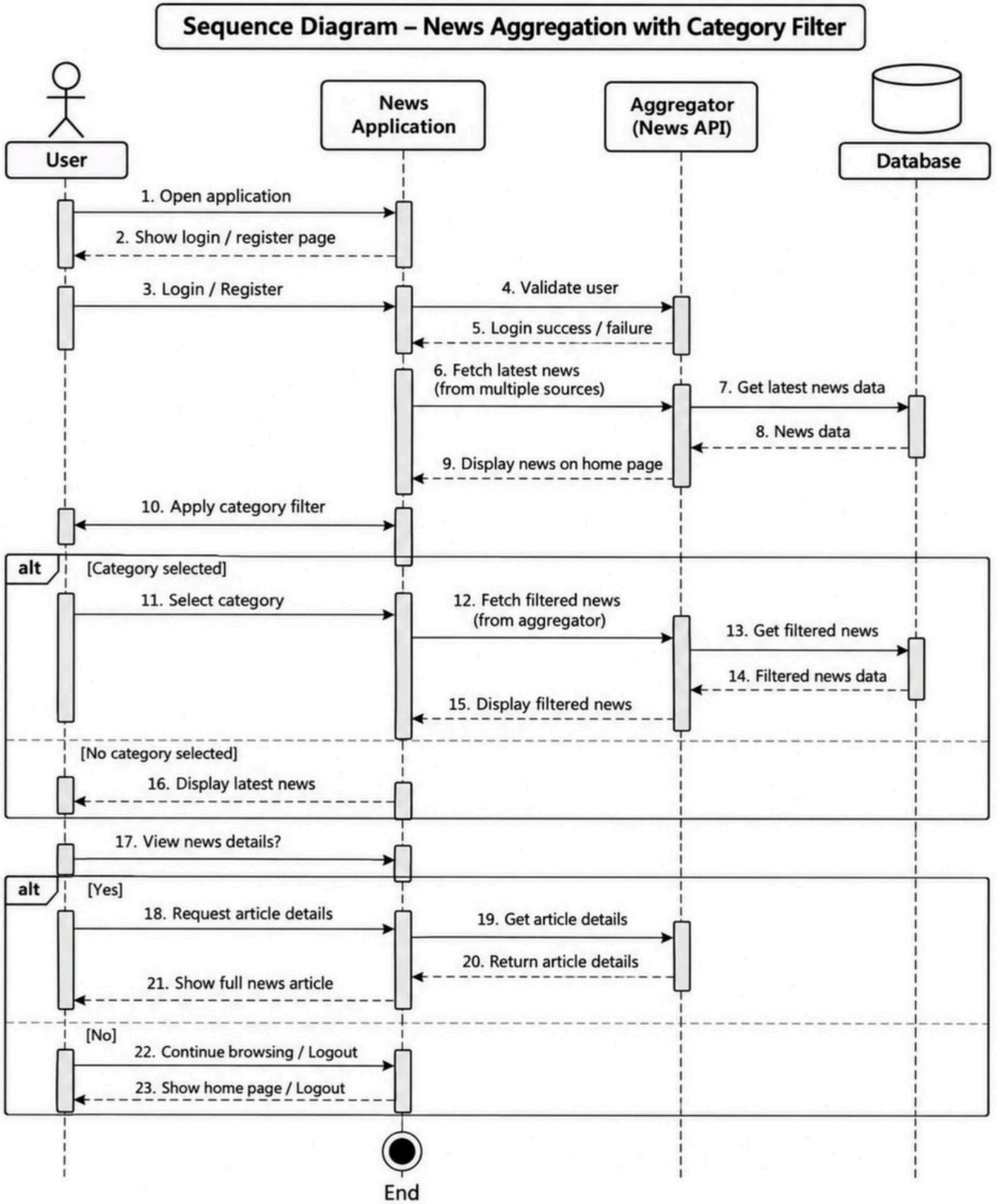
1. **Descriptive Analysis:**
   - Shows the available news articles based on categories such as Sports, Politics, Technology, Business, Entertainment, Health, etc.

2. **Diagnostic Analysis:**
   - Analyzes why certain news is popular, frequently viewed, or receiving more attention from users.

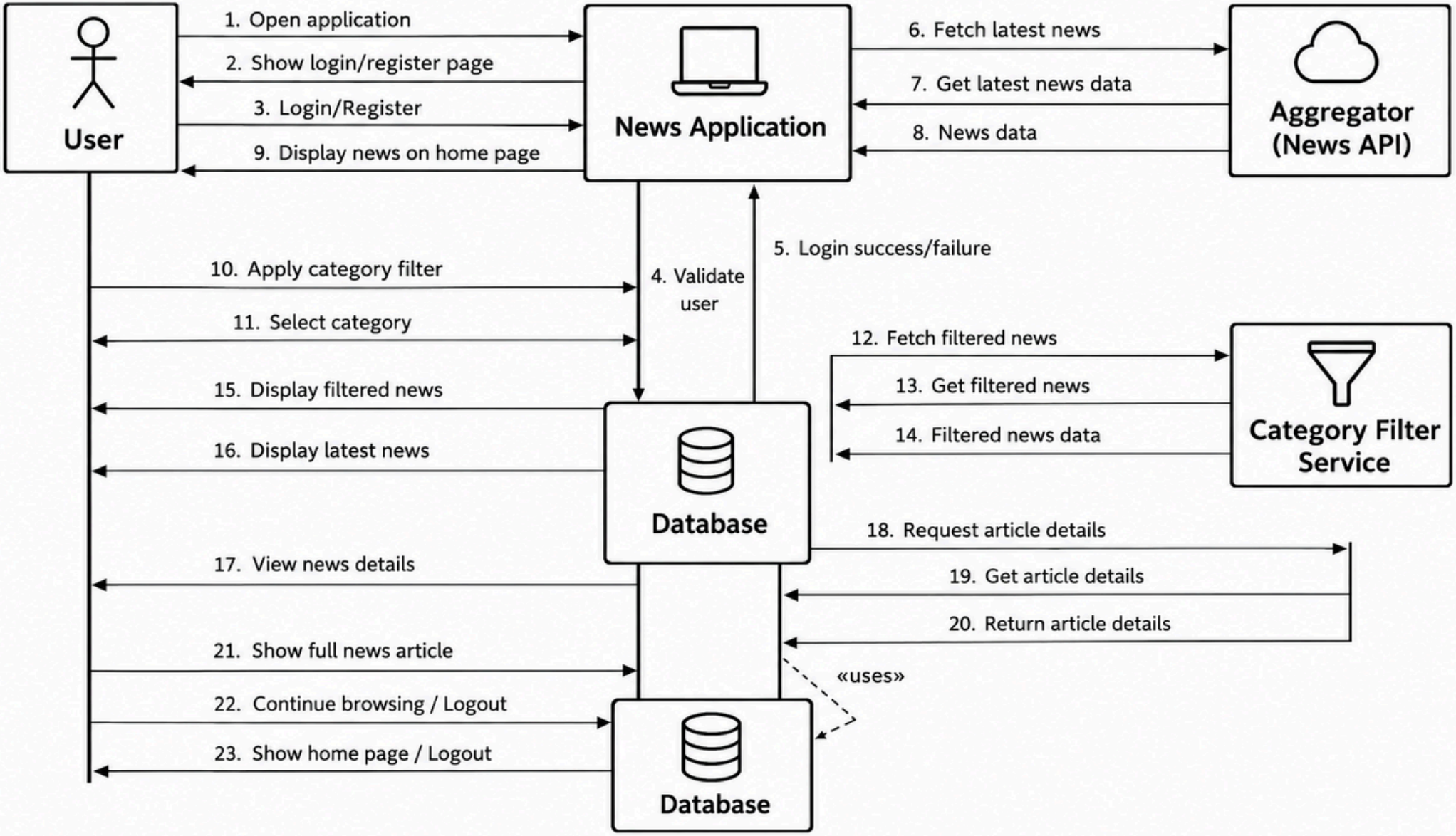
3. **Prescriptive Analysis:**
   - Recommends relevant news to the user based on their selected category and reading preferences.

4. **Predictive Analysis:**
   - Predicts which news topics or categories may become popular or trending based on previous user activity and news trends.

# Sequence Diagram – News Aggregation with Category Filter



# Collaboration Diagram – News Aggregation with Category



# Class Diagram – News Aggregation with Category Filter

